

TASK RECORD

NAME: VIKASINI.K

DEPARTMENT: B. Tech CSE (Artificial Intelligence and Data Science)

COLLEGE: Vels institute of science, Technology and Advanced studies

INTERNSHIP ROLE: Data Analyst

TASK: DIFFERENCE BETWEEN AGILE AND WATERFALL ALONG WITH THEIR USE CASES

Task Description:

To understand the differences between the Agile and Waterfall software development methodologies and identify their suitable use cases.

INTRODUCTION:

In Software Development Life Cycle (SDLC) is a process used to develop software efficiently. Agile and Waterfall are two popular SDLC methodologies. They differ in planning, development, testing, and project management.

AGILE METHODOLOGY:

Agile is an iterative software development methodology where the project is divided into small development cycles called sprints. Teams work closely with customers, receive continuous feedback, and improve the product throughout the development process.

Agile is an iterative approach that emphasizes flexibility, collaboration, and continuous delivery.

KEY FEATURES:

- Iterative development
- Frequent customer feedback
- Continuous testing
- Quick delivery
- Flexible to changing requirements

WATERFALL METHODOLOGY:

Waterfall is a traditional software development methodology where each phase is completed before moving to the next. The phases typically include Requirement Analysis, Design, Development, Testing, Deployment, and Maintenance.

Waterfall follows a sequential approach where each phase is completed before moving to the next. Selecting the appropriate methodology depends on the project requirements, timeline, and business objectives.

KEY FEATURES:

- Sequential process
- Detailed documentation
- Fixed requirements
- Structured planning
- Easy project tracking

DIFFERENCE BETWEEN AGILE AND WATERFALL:

FEATURE	AGILE	WATERFALL
1. Development Model	Iterative and Incremental	Sequential and Linear
2. Flexibility	High	Low
3. Requirement Changes	Easily accepted	Difficult after planning
4. Customer Involvement	Continuous	Limited
5. Testing	During every sprint	After development
6. Delivery	Frequent releases	Single final release

7. Documentation	Minimal	Extensive
8. Risk	Lower	Higher
9. Time to Deliver	Faster	Longer

USE CASES OF AGILE:

Agile is suitable for:

- Mobile Application Development
- Web Application Development
- Artificial Intelligence Projects
- Machine Learning Projects
- Startup Products
- E-commerce Websites
- SaaS (Software as a Service) Applications

USE CASES OF WATERFALL:

Waterfall is suitable for:

- Banking Systems
- Government Projects
- Hospital Management Systems
- Construction Management Software
- Manufacturing Systems
- Embedded Systems
- Defense Projects

AGILE:

Advantages :

- Faster product delivery
- Continuous customer feedback
- Easy to accommodate changes
- Improved software quality

Disadvantages:

- Less predictable cost and timeline
- Requires active customer involvement
- Documentation may be limited

WATERFALL:

Advantages:

- Simple and easy to understand
- Well-defined documentation
- Easy project management
- Suitable for stable requirements

Disadvantages:

- Difficult to make changes later
- Testing starts only after development
- Higher risk if requirements change

CONCLUSION:

Both Agile and Waterfall are effective SDLC methodologies with different strengths. Agile is best suited for projects where requirements evolve over time and rapid delivery is important. Waterfall is more appropriate for projects with fixed requirements, clear documentation, and strict regulatory standards. Choosing the right methodology helps improve project quality, customer satisfaction, and overall development efficiency.

